

Cleaning Schedule Add-on — Plain English Overview

2026-05-06 · Version 1.17.x · For reference and planning conversations

1. What Is This?

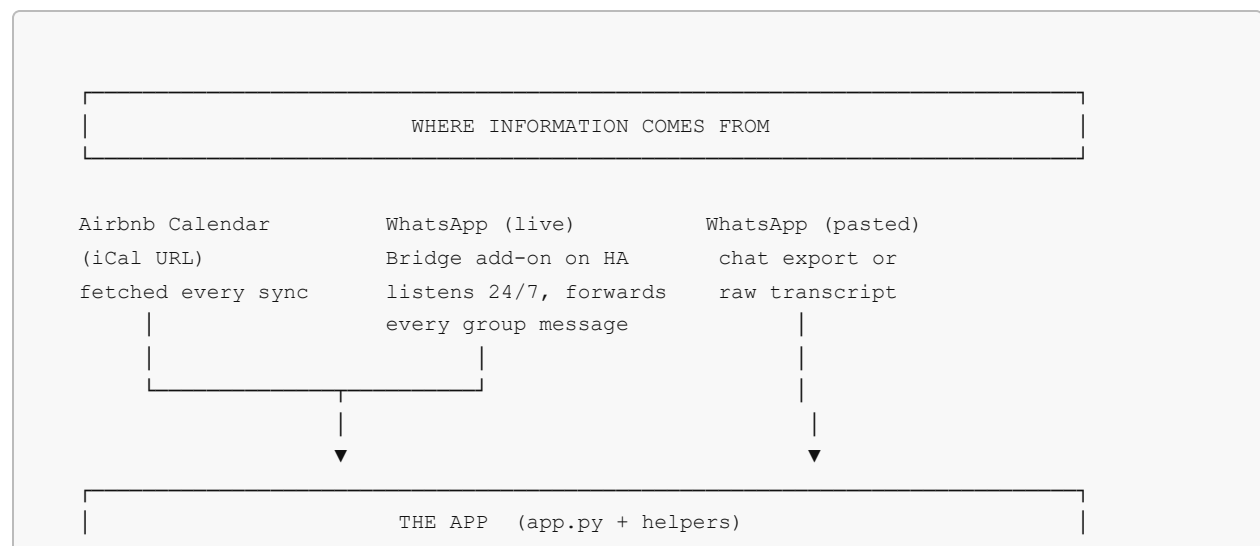
This is a small web app that runs on your Home Assistant server at home. Its job is to keep track of which cleaner is scheduled for each Airbnb checkout, and to stay in sync with whatever you and the cleaners have discussed over WhatsApp.

The app pulls your Airbnb booking calendar automatically, lets you assign cleaners to each checkout, and then watches the WhatsApp group chats to detect when a cleaner has confirmed or declined. It uses Claude (an AI) to read those WhatsApp messages and figure out what was agreed.

Everything it knows is saved to one file on the server. Google Calendar is used purely as a display layer — so Michelle and the cleaners can see upcoming jobs without needing to log into anything. The app is the brain; Google Calendar is the whiteboard.

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2. The Big Picture



- When a booking arrives:

- add it to the list

- mark old ones cancelled

- flag unassigned ones for review
- When a WhatsApp message arrives:

- Claude reads it (two separate calls)

- "Parse": which booking does this relate to, and what's the action?

- "Facts": what scheduling claims does this message make at all?

- High confidence + known cleaner → apply automatically

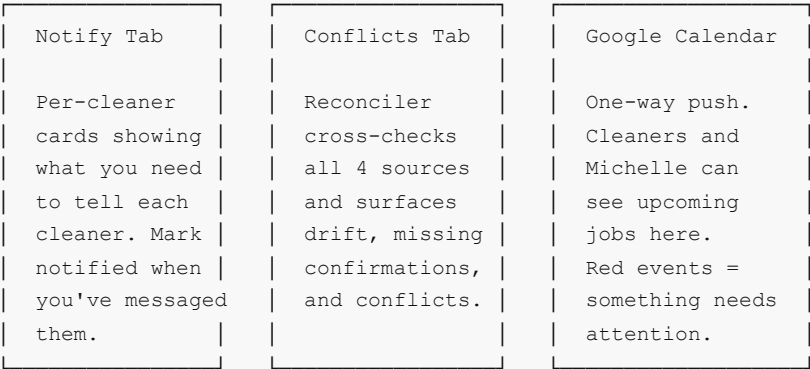
- Anything else → Review queue

data.json

(the single truth file)

bookings + messages

+ facts + cleaner map



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3. The Home Page (Three Tabs)

Notify Tab — the main daily view

This is the tab you use most. It shows one cleaner at a time, and lists every job where what you've told that cleaner doesn't match what's currently booked. For example: if you told Maria she was cleaning on the 15th, but the booking got cancelled — she shows up here because you need to tell her.

The possible situations are:

- **New** — you've assigned a cleaner but haven't told them yet.
- **Changed** — the date or time has shifted since you last communicated.
- **Cancelled** — the booking was cancelled after you'd already told the cleaner.
- **Unassigned** — there's an upcoming checkout with no cleaner assigned at all.

When you've sent the message to that cleaner, hit "Mark notified" and the app records what they were told. The card disappears until something changes again. You move through cleaners one at a time using the arrow buttons.

WhatsApp Review Tab

Messages that Claude wasn't confident enough to apply automatically end up here. You can read each one, then accept Claude's suggestion, override it with the correct booking and cleaner, or ignore the message entirely. You can also name groups here (so "123456789@g.us" shows as "Maria group") and map unknown phone numbers to cleaners.

Conflicts Tab

The reconciler runs in the background and cross-checks all four data sources — Airbnb calendar, your local bookings, Google Calendar, and what cleaners have actually said on WhatsApp. If something doesn't add up, it surfaces here as a finding. Findings are ranked by urgency. You can take action directly (assign a cleaner, open the booking to edit) or dismiss a finding if you know it's resolved some other way.

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4. How Bookings Work

Airbnb bookings (automatic)

When you press "Sync" on the home page, the app fetches your Airbnb calendar URL and imports any new reservations. If a reservation disappears from Airbnb (cancelled by the guest), it gets marked cancelled in the app too. Everything you've added by hand — custom stays, manual cleanings — is left alone by the sync.

Adding things by hand

The Add page lets you create two kinds of entries manually:

- **Stay** — a guest stay with a start and end date. You'd use this for non-Airbnb bookings.
- **Cleaning** — a standalone cleaning job on a specific date. Good for deep cleans or jobs not tied to a guest checkout.

The booking edit page

From here you can change the assigned cleaner, set or update the cleaning time, and add notes. For cancelled bookings, there's a Dismiss button to remove it from view. Airbnb bookings that are still active can only have their cleaner and time changed — they can't be deleted (Airbnb decides when they get cancelled).

The print view

A clean month-by-month table showing stays as coloured bars and cleaner assignments on checkout days. Designed to look good on paper. Michelle uses this.

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5. How WhatsApp Works

The WhatsApp Bridge (live monitoring)

A second add-on running on Home Assistant. It logs into WhatsApp as a linked device (like WhatsApp Web on a phone). Once paired by scanning a QR code once, it stays connected and forwards every message from the specified group chats to the main app. It never sends anything — read-only.

Both your own outbound messages and the cleaners' replies are forwarded. This matters because your messages provide scheduling context for Claude to understand the replies.

What happens when a message arrives

Claude reads each message twice, in parallel:

- **Parse pass** — "Does this message refer to a specific booking I know about, and if so, is the cleaner confirming, declining, or proposing a time change?" If confidence is high and the cleaner is recognised, the booking gets updated automatically.
- **Facts pass** — "What scheduling claims does this message make, regardless of whether I can match them to a specific booking?" This feeds the reconciler so it can detect things like "a cleaner said yes but the booking still shows unassigned."

Messages that Claude can't confidently act on go to the Review tab for you to handle manually.

Backfill — catching up on past chats

If you've already been running the property for a while before installing this app, you can export a WhatsApp chat (from your phone: open the group → three dots → More → Export chat → Without media) and paste it in. Claude will read through it and suggest which cleaner to assign to each unassigned booking, with a confidence score. You review the suggestions and apply the ones that look right.

Transcript ingest — for historical data

Similar to backfill, but instead of matching to unassigned bookings, it parses the transcript into the full message log and runs the facts extraction on each message. This gives the reconciler full historical context. Useful for a one-time catch-up when you first install. Re-running the same transcript is safe — duplicate messages are ignored.

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6. The AI Layer (Claude Haiku)

The app uses Claude Haiku — a small, fast, cheap AI model — for two distinct jobs. It never makes decisions on its own; it only extracts structured information from unstructured WhatsApp text and hands that back to the app to act on.

Job 1: Parse (route one message to one booking)

Given a message, its recent context, and the list of upcoming bookings, Claude answers: "Which booking does this refer to? Which cleaner? What action?" It returns a confidence score. Below 85% confidence, or if it can't match a known cleaner or booking, the message goes to the Review tab.

Job 2: Facts (extract all scheduling claims)

Given the same message and context, Claude also answers a broader question: "List every date, cleaner, and time claim in this message." A single message like a re-posted schedule with 15 dates might produce 15 separate facts. These are stored separately and used by the reconciler to cross-check everything.

The context window for this pass is capped at the last 30 messages from the same group — a practical limit to prevent runaway API costs during bulk imports.

Prompt versioning

The facts prompt has a version stamp. If you change the prompt (to improve accuracy or fix a pattern it's getting wrong), you increment the version number and run a reprocess job from the admin panel. This re-reads all stored messages with the new prompt. Old and new records coexist safely until reprocessing is complete.

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7. The Reconciler (Conflict Detection)

The reconciler is the app's auditor. It compares all four sources of truth against each other and flags anything that doesn't line up. You run it manually with the "Run digest" button, or it runs automatically when you load the Conflicts tab.

What it checks

- **Airbnb calendar vs. your bookings** — is there a reservation in Airbnb that you don't have a booking for? Or a booking in your records that disappeared from Airbnb?
- **Your bookings vs. Google Calendar** — is GCal out of date? Missing events? Events that no longer correspond to a booking?
- **Notify queue drift** — are there bookings where what you've told a cleaner no longer matches reality?
- **WhatsApp facts vs. bookings** — has a cleaner confirmed something that your records don't show? Is someone assigned to a job they declined on WhatsApp? Did two cleaners both confirm the same date?
- **Cleaner change-of-mind** — did a cleaner first confirm and then later decline (or vice versa) for the same date?
- **Host schedule vs. bookings** — did you post a schedule in the chat that names a different cleaner than what's recorded?

Urgency levels

- **Needs attention** — something is actively wrong and should be resolved.
- **Suggestion** — something looks like it might be incomplete (e.g. an unrecorded confirmation).
- **Informational** — interesting context but no immediate action required.

Dismissing findings

If you know a finding is resolved or not relevant, you can dismiss it from the Conflicts tab. It stays dismissed even if you re-run the reconciler. You can undo a dismissal if needed.

Known noise

Old messages where you (as host) posted a schedule in the group chat can generate "schedule mismatch" findings even after those jobs are long done. This is a known limitation — the reconciler doesn't yet filter out stale host messages. Until fixed, expect some noise from older backfilled transcripts.

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8. Google Calendar

The app pushes a one-way copy of the schedule to a Google Calendar shared with Michelle and the cleaners. No one edits the Google Calendar — it's purely for viewing. All changes go through the app and get pushed to GCal automatically.

What the calendar shows

- Each cleaning job appears as a timed event (default 11 AM – 1 PM if no time set, or from the clean time for 2 hours).
- Cleaner's name is in the event title. Colour corresponds to the cleaner (each cleaner gets a consistent colour).
- **Red events (⚠)** — something needs attention: either the cleaner hasn't been told yet, or what they were told is no longer current.
- If a booking is cancelled after the cleaner was notified, a red " ❌ Cleaning Cancelled — Notify [name]" event stays on the calendar until you mark that cleaner as notified.

Setup

Requires a Google Cloud service account — a special set of API credentials that doesn't need a login or refresh tokens. You create it once, share the target calendar with the service account's email address, and paste the credentials into the add-on settings. The `gcal_auth.py` script validates the credentials and tells you the exact email to share with.

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9. Under the Hood (for context, not operation)

Where data lives

One JSON file on the server: `/data/data.json`. It contains the bookings list, all WhatsApp messages, all extracted facts, the cleaner-to-phone-number mapping, and the group chat labels. There's no database. The file persists across app restarts and updates.

How it runs

A Python web app (Flask) running inside a Home Assistant add-on container. The app is a single file (~3,000 lines). The WhatsApp Bridge is a separate Node.js add-on running alongside it. They communicate over the local network.

Auth and access

The app is accessible through the Home Assistant sidebar (via HA's ingress proxy, which handles login). It's also reachable directly on port 5000 on your local network for tooling purposes — those external calls need to include a shared secret in the header. The WhatsApp Bridge calls over loopback (same machine) and doesn't need the secret.

Updating the app

Push the code change to GitHub, bump the version number in the config file, then refresh the add-on in Home Assistant. HA won't pick up changes unless the version number changes — this is a HA limitation. You can't edit files in place via the Windows file share (SMB caching causes silent failures) — always go through GitHub.

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10. Things Not Done Yet

What	Why it's not done / what's blocking it
Daily digest notification	The reconciler produces too much noise from old host messages. Fix the noise first, then the digest is useful.
Stale host message noise	When you posted a schedule in WhatsApp months ago, that old message still generates conflict findings today. The reconciler needs to learn to ignore superseded host messages.
Duplicate fact assertions	If you said "Itzel, May 19" in two different messages, that's currently two separate facts instead of one. Not painful yet, so deferred.
Jokey/narrative host messages	Claude sometimes over-reads casual messages from you in the group chat as scheduling assertions. Fix: mark the host's phone number in config so those messages are interpreted differently.
First-run notify queue flood	When first installed, every assigned booking looks "new" because there's no record of it having been communicated. Need a one-time "trust what's already here" button.
Dedicated bot WhatsApp account	Currently using a personal phone number. Moving to a dedicated SpeakOut number (~\$125/yr) would be cleaner. The re-pair procedure is documented but not done yet.
Automated UI tests	The notify queue, Conflicts tab, WhatsApp review flow, and backfill pages aren't covered by end-to-end tests yet.

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11. General Notes

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